

SMART FOOD SPOILAGE PREDICTION AND ALERT SYSTEM

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ABSTRACT

Food spoilage leads to significant waste and health hazards. This project presents a Smart Food Spoilage Detection System that utilizes the Internet of Things (IoT) and Artificial Intelligence (AI) to monitor food freshness in real-time. The system is built using an ESP32 microcontroller integrated with gas and environmental sensors (MQ-135, MQ-5, MQ-6, DHT22) to detect spoilage indicators such as ammonia (NH₃), carbon dioxide (CO₂), temperature, humidity, and other environmental gases. The system detects spoilage when gas concentrations exceed safe thresholds, such as NH₃ > 10 ppm, CO₂ > 1000 ppm, and volatile organic compounds (VOCs) > 400 ppm. Collected sensor data is transmitted via Wi-Fi/Bluetooth to a cloud platform, where a machine learning model analyzes patterns and predicts spoilage. Users can monitor real-time data through a mobile application dashboard, enabling proactive food management. This cost-effective and scalable solution enhances food safety, reduces waste, and improves storage efficiency for households and industries.

Keywords : Food Spoilage, Iot, Gas Sensors, Real Time Monitoring, ESP32.

1. INTRODUCTION

Food spoilage detection has traditionally depended on human judgment, relying on sensory indicators like appearance, smell, and taste. However, these methods are prone to error and inconsistencies, leading to the possibility of premature disposal of food items or consumption of spoiled food, both of which pose risks to health and waste management. To address these concerns, there is a growing need for automated, real-time systems that can detect food spoilage with a high degree of accuracy. This research introduces an IoT-based system designed to monitor and detect food spoilage in real-time using a combination of environmental and chemical parameters. The system uses a variety of sensors to track key indicators such as ammonia (NH₃), carbon dioxide (CO₂), temperature, humidity, and other environmental gases all of which are known to be closely linked with the degradation of food quality. This paper aims to evaluate the effectiveness of this IoT-based solution, compare it with traditional spoilage detection methods, and discuss its potential impact on food safety, waste reduction, and the broader food industry.

2. LITERATURE REVIEW

Sanjit Manohar Majhi et al. (2021) investigated the application of chemo resistive sensors in food spoilage detection. These sensors possess a self-heating capability that aids in power consumption optimization. However, a major drawback of this approach is the dependence on external heaters for effective operation, which could increase overall energy consumption and system complexity.[1]

Dr. P. Srinivasu et al. (2024) proposed an IoT-based food spoilage monitoring system that enhances decision-making by providing real-time data. However, the study suggests that the system's accuracy and efficiency can be further improved by incorporating additional sensors to capture a wider range of spoilage indicators.[2]

Disha Bongarde et al. (2024) explored the integration of Artificial Intelligence (AI) and Machine Learning (ML) within IoT-based monitoring systems. Their research focused on real-time monitoring of food spoilage using AI-driven predictive analysis. Despite its advantages, the study identified challenges in the seamless integration of AI and ML, which require further refinement to enhance the system's performance and reliability.[3]

The Artificial Neural Network data is trained using Stochastic Gradient Descent and the Back Propagation technique. The sensor chamber is positioned underneath the tomato-based Filipino food for simple detection of the gas sensors generated by the cuisine. The research might be beneficial in determining whether or not food has gone bad. This electronic nose device may detect the degree of rotting in a certain tomato-based Filipino dish. The researchers assigned rotting levels ranging from 0 to 12. Starting at 7:00 a.m. on Day 1, these levels correlate to the period when the meal is monitored every 4 h. As the trial progressed, the food deteriorated often between levels 5 and 6. The electronic nose system has a 3.85% mistake rate, according to the confusion matrix. [4, 5] Food is extremely essential in our day-to-day lives. The quality of food is deteriorating day by day as globalization progresses. To keep the food fresh, various food processing is done most of the time. Various preservatives or substances are added to the meal to give it a fresh or appealing appearance. The majority of food is now preserved using chemicals, resulting in food contamination. As a result of the pollution, numerous illnesses develop, prompting consumers to seek out healthier foods. Organic food is desired by the public in order to maintain a healthy lifestyle. So, in order to prevent difficulties linked with food that cannot be interpreted by humans, we need a gadget that can assist in determining the quality of food. There is a need for such a device that instructs us on how to eat sanitary meals. As a result, in order to meet this customer need, we created a gadget that determines if the food quality is good or terrible. This article examines the application of several sensors in the food business. Sensors such as pH sensors, gas sensors, and temperature sensors aid in determining the state of food. This system has a strong presence in restaurants, homes, and small businesses [6].

When food is reversed in an inadvertent situation, such as an insufficient temperature, foodstuff rots as a result of the rapid breeding of food spoilage bacteria in warm and wet conditions where bacteria can easily be generated. The consequences can be even worse if humans eat those degenerative foodstuffs, as they may develop bromatoxism. The goal of this study is to design and simulate a wearable RFID patch for food spoilage monitoring with smart packaging that can be recognized and read temperature information by a device that supports near NFC technology via an attached circular antenna.[7]

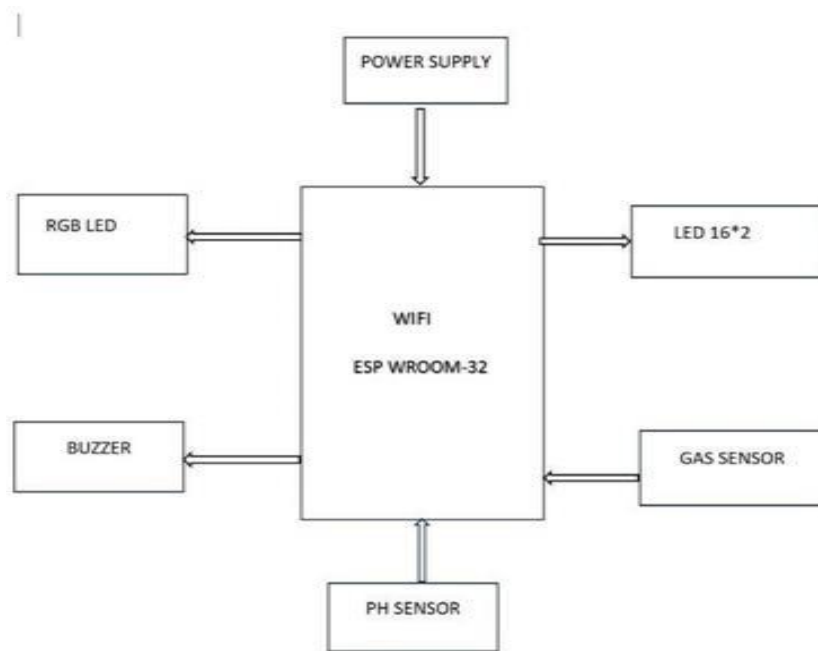
Microbial deterioration in milk is a major source of concern for those who want to eat healthily. The goal of this research is to use an electronic nose system (e-nose) with a variety of gas sensors as an initial tool for monitoring food deterioration in order to preserve food safety and human health. Nanoparticles gas sensors made of various polymers and f-SWCNTs are used in this device. When these gas sensors are exposed to the headspace of the milk, they respond to the VOCs present in the milk. Pasteurized milk was stored at 4 and 25°C for 9 days to monitor the development of spoiling. The odor pattern was examined using PCA to assess the capabilities of an electronic nose for freshness and milk spoiling detection. Monitoring the change in detecting responses revealed each sensor's capacity to identify the strength of the odor level. The findings of the discrimination demonstrate that the odor levels of samples maintained at 4°C do not change from the first to the ninth day, but the odor levels of samples held at ambient temperature increase with storage duration.[8]

Food planning and quality assessments are becoming increasingly essential aspects of modern society. We may effectively analyze crop quality utilizing various learning approaches thanks to the scientific vision and agricultural advancements, which certainly replace inconsistent, unpredictable, and time-consuming human labor. The use of fuzzy logic to crop quality has effectively demonstrated its worth and established itself as a viable alternative to automation, indicating that it has a bright future ahead of it. This article examines some of the most important research findings on this subject.[9]

The thermal stability of the food products during the transportation was studied and was calculated for a 24 hr basis. However, real time monitoring can be implemented for the up to time calculations.[10]

3. EXISTING SYSTEM

The current food spoilage detection systems use an ESP32 microcontroller along with MQ-135 gas sensors and pH sensors. These systems typically rely on simple threshold- based alerts, where an LED or buzzer activates when gas levels exceed a predefined limit. While these systems provide a basic level of spoilage detection, they face several limitations:



Limited Sensor Diversity: Most existing systems use a single type of gas sensor, reducing detection accuracy for different spoilage indicators.

Lack of Predictive Analytics: Existing systems rely on immediate threshold breaches rather than AI- driven pattern analysis, limiting early warnings.

Short Sensor Lifespan: Some sensors degrade over time, leading to inaccurate readings without periodic calibration.

Lack of Remote Monitoring: Most traditional systems require physical presence to checks poilage alerts, making them inefficient for industrial applications.

Our research addresses these limitations by integrating multiple sensor types and machine learning algorithms for enhanced accuracy and predictive capabilities.

4. PROPOSED SYSTEM

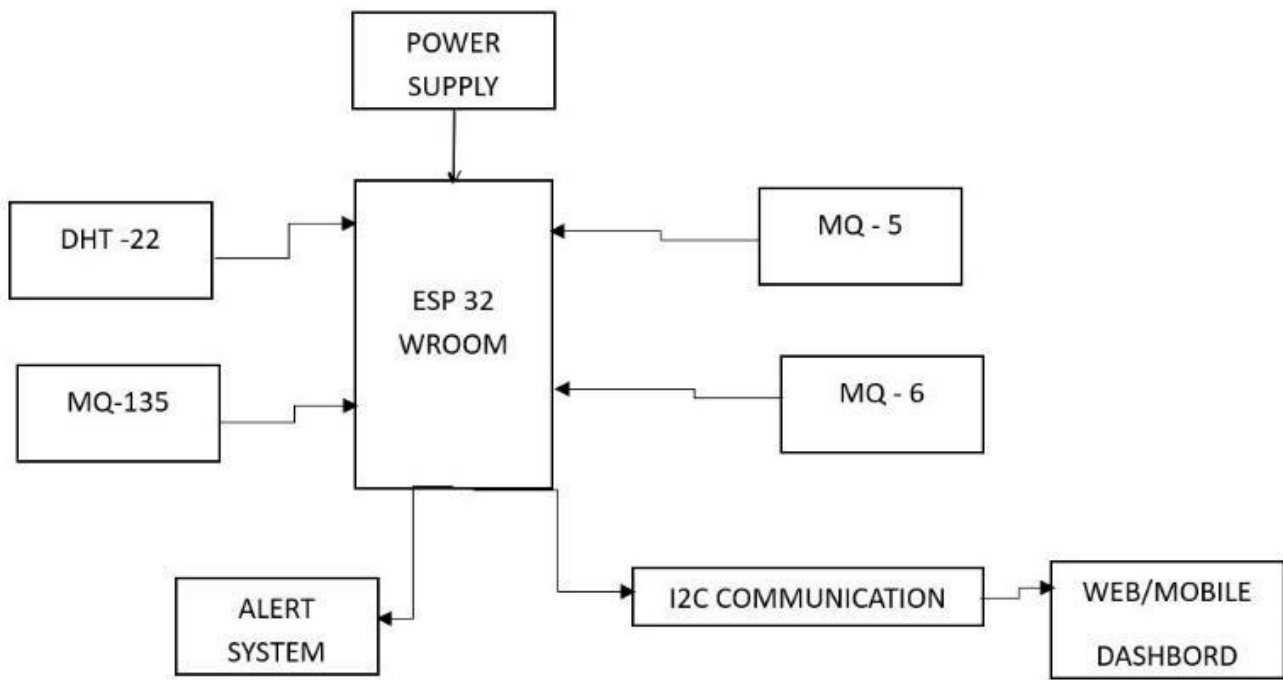
The proposed system enhances accuracy and sensor longevity by integrating multiple gas and environmental sensors.

Spoilage alerts are transmitted through Wi-Fi modules in the ESP32 using real-time data analysis. The hybrid sensor approach improves detection range and prediction accuracy.

4.1 Stem Architecture

4.1.1 The system consists of three main components:

Sensing Unit: Includes MQ-135 (air quality), DHT22 (temperature & humidity), and a alcohol sensor. This unit continuously collects food storage data.



4.1.2 Processing Unit:

The ESP32 microcontroller processes sensor readings and transmits them via Wi-Fi/Bluetooth to a cloud server.

4.1.3 User Interface:

A mobile dashboard provides real- time monitoring and alert notifications to users.

5. DESIGN METHODOLOGY

Step1: Problem Definition

- Objective: Detect food spoilage using gas sensors and environmental factors.
- Inputs:
 - MQ-135→Airquality(CO₂, ammonia, etc.)
 - MQ-5→LPGandmethane gases
 - MQ-6 → Alcohol and hydrocarbon gases
 - DHT Sensor → Temperature and humidity
- Output: "Fresh" or "Spoiled" displayed on LCD and sent to a web dashboard.

Step2: Hardware Design

Components Used

- ESP32:MicrocontrollerwithWiFiforIoT connectivity
- Sensors:MQ-135,MQ-5,MQ-6,DHT22
- (Humidity & Temperature)
- LCDDisplay:16x2I2CLCDforreal-time status
- Buzzer and LED: Alerts when food is detected as spoiled

STEP3: Sensor Connections

- MQ-135 (Analog) GPIO34
- MQ-5 (Analog) GPIO35
- MQ-6 (Analog) GPIO32
- DHT22 (Digital) GPIO4
- LCD I2C GPIO21(SDA),
GPIO22(SCL)
- Buzzer (Digital) GPIO27

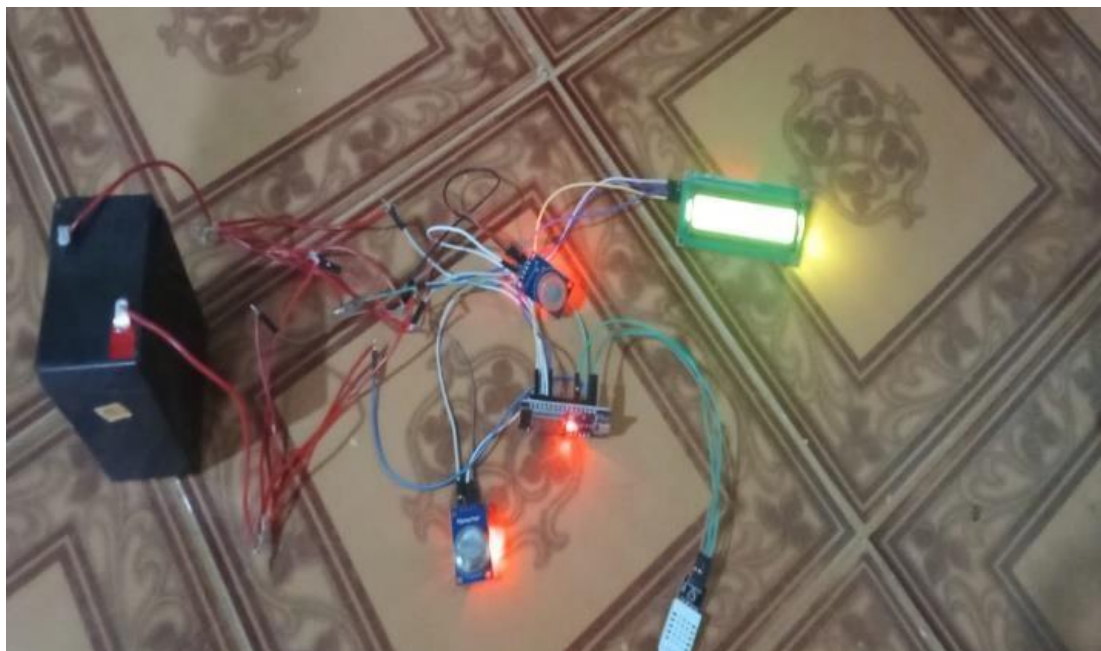
STEP4 : Communication & IoT Integration

- ESP32 sends data to Flask server over WiFi
- LCD displays "Fresh" or "Spoiled"
- Buzzer alerts if food is spoiled

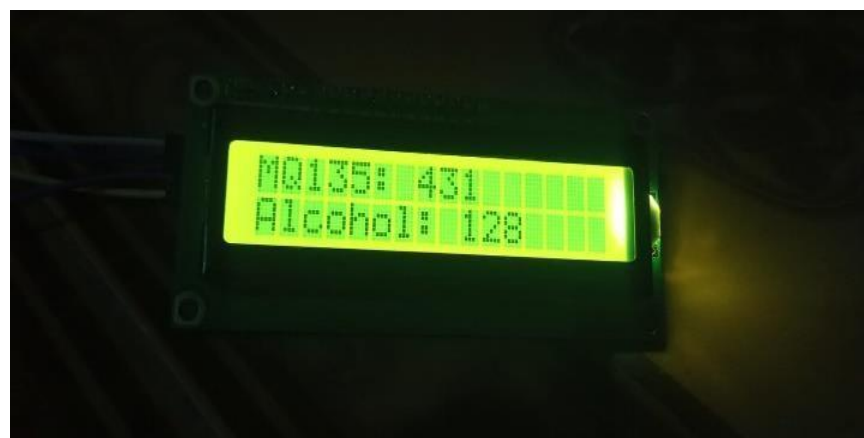
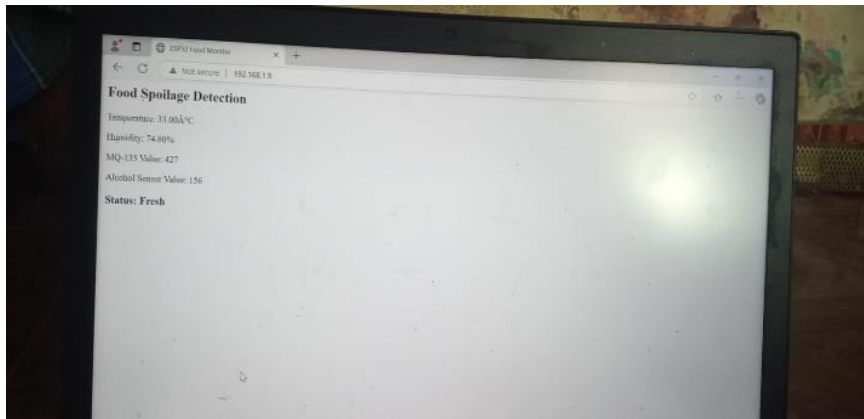
STEP5: Display & Alerts

- LCD shows "Fresh" or "Spoiled"
- Buzzer beeps if food is spoiled
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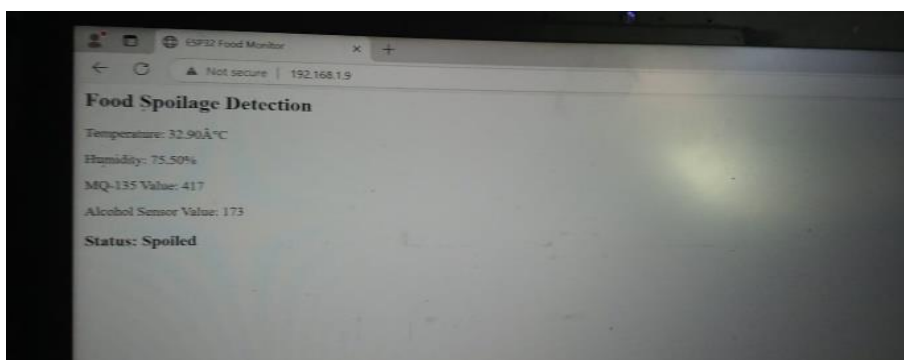
6. RESULTS



6.1 Displaying Fresh Data:



6.2 Displaying of Spoiled Data:



7. FUTURESCOPE

1. AI Optimization: Train a better model with real-world food spoilage datasets
2. Edge AI Processing: Convert model to Tensor Flow Lite for ESP32
3. Automated Actions: Trigger cooling systems if spoilage is detected
4. Voice Assistant Integration
5. Low-Power Sensor Modes: Use deepsleep modes for ESP32 to conserve battery when monitoring.

8. CONCLUSIONS

The ESP 32 is connected to the multiple sensor and the output data of the various sensors is displayed through the led and also the real time data is monitored by the web dashboard through I2C communication of the ESP 32 module. Hence the food spoilage is predicted and information is displayed and sent to the user by using the lcd through I2C communication protocol and the alert is given by using the led and buzzer systems.

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